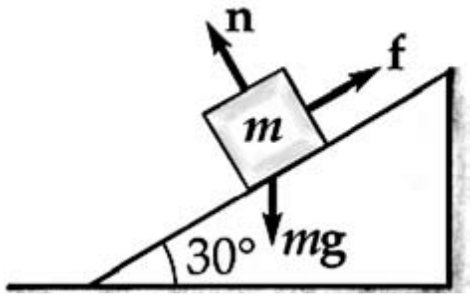
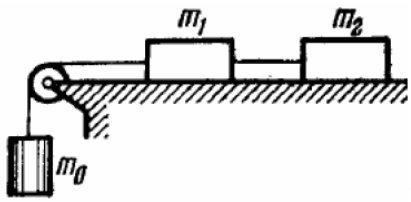
PHY101: Introduction to Physics-1

Monsoon, 2017

Tutorial-7

1. A 3 kg block, as shown in figure, starts from rest at the top of a 30° incline and slides a distance of 2 m down the incline in 1.5 s. Find (a) the magnitude of the acceleration of the block, (b) the coefficient of kinetic friction between block and plane, (c) the friction force acting on the block, and (d) the speed of the block after it has slid 2 m.



2. In the arrangement of figure below the masses m_0, $m_1$and m_2 of bodies are equal, the masses of the pulley and the threads are negligible, and there is no friction in the pulley. Find the acceleration a with which the body m_0 comes down, and the tension of the thread binding together the bodies m_1 and m_2 if the coefficient of friction between these bodies and the horizontal surface is equal to k.